Assignment 1: Deep Learning

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***Abstract*—This document outlines the methodology and model selection process for predicting time-series data using neural networks in Deep Learning Assignment 1.**

***Index Terms*—Deep Learning, Neural Networks, Time Series Forecasting**

I. Introduction

Assignment:

(a) Select your choice of neural networks model that is suitable for this task and motivate it. Train your model to predict one step ahead data point, during training (see following Figure). Scale your data before training and scale them back to be able to compare your predictions with real measurements.

(b) How many past time steps should you input into your network to achieve the best possible performance? (Hint: This is a tunable parameter and needs to be tuned).

(c) Once your model is trained, use it to predict the next 200 data points recursively. This means feeding each prediction back into the model to generate the subsequent predictions.

(d) On May 8th, download the real test dataset and evaluate your model by reporting both the Mean Absolute Error (MAE) and Mean Squared Error (MSE) between its predictions and the actual test values. Add

II. Used Models

We have chosen to implement the [Insert Name of Neural Network Model] for our deep learning assignment. This model is well-suited for [briefly explain why this model is appropriate for the task at hand]. The architecture of this model allows it to effectively capture [mention specific features or patterns relevant to the task], making it a strong candidate for achieving high performance on our dataset.

III. Approach

IV. Results

V. Conclusion

References